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Statistical Estimation: MLE, MAP, Full Bayesian

MLE

If India & Australia plays tomorrow, will India win or lose?

Past data: Win, Loss, Loss, Win, Win

So,
$$p(\text{India Wins}) = \frac{3}{5}$$

Now we'll fit a statistical model & estimate MLE, and see whether we can obtain the same answer.

There can be two outcomes: $\{ \text{Win, Loss} \}$
View this as a random variable with 2 states:

$$Y = \{0, 1\}$$

Let the hypothesis class is:

$$Y \sim \text{Bernoulli}(\theta)$$

$$Y = \begin{cases} 1, & \text{prob } \theta \\ 0, & \text{prob. } (1-\theta) \end{cases}$$

↓
the estimation problem is to find θ .

The dataset is: $D = \{1, 0, 0, 1, 1\}$

Some trivial choices: $\begin{cases} \hat{\theta} = 0 \\ \hat{\theta} = 1 \end{cases}$ bad estimators.

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \underbrace{p(D|\theta)}_{\text{Likelihood}}$$

[Likelihood is a function of the parameters θ]

- Let $D = \{1\} \Rightarrow L(\theta) = p(Y=1|\theta) = \theta$
- $D = \{1, 0\} \Rightarrow L(\theta) = p(Y=1|\theta) p(Y=0|\theta) = \theta(1-\theta)$
- $D = \{1, 1, 1\} \Rightarrow L(\theta) = \theta^3$

[I.I.D. samples]
(independency)

$x_H \rightarrow \# \text{ of wins}$
 $x_T \rightarrow \# \text{ of losses}$

In general:
$$L(\theta) = \theta^{x_H} (1-\theta)^{x_T}$$

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta)$$

[monotonic f_h]

$$\Rightarrow \hat{\theta}_{MLE} = \arg \max_{\theta} [\alpha_H \log \theta + \alpha_T \log (1-\theta)]$$

$$\frac{d}{d\theta} [\alpha_H \log \theta + \alpha_T \log (1-\theta)] = 0$$

$$\Rightarrow \frac{\alpha_H}{\theta} - \frac{\alpha_T}{1-\theta} = 0$$

(assume $\theta \neq 1, \neq 0$)

$$\Rightarrow \hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$

So in our dataset, $\alpha_H = 3, \alpha_T = 2$

Hence, $\boxed{\hat{\theta}_{MLE} = \frac{3}{5}}$

(matches with our previous estimation)

Why Max-Likelihood?

Leads to "natural" estimators

MLE is optimal if model-Class (hypothesis) is correct.

Let Y is a discrete random variable:

$$Y = y \in \{1, \dots, K\}$$

Consider log-Likelihood (for some model) average:

$$\frac{1}{N} LL(\theta) = \frac{1}{N} \sum_{i=1}^N \log P(y_i | \theta)$$

$$= \frac{1}{N} [\#(Y=1) \log P(Y=1|\theta) + \#(Y=2) \log P(Y=2|\theta) + \dots]$$

$$\begin{aligned} &= P^*(1) \log P_{\theta}(1) + \dots \\ &= \sum_{y=1}^K P^*(y) \log P_{\theta}(y) = \sum_y P^*(y) \log \left[P_{\theta}(y) \frac{P^*(y)}{P^*(y)} \right] \\ &= \sum_y P^*(y) \log P^*(y) - \sum_y P^*(y) \log \left[\frac{P^*(y)}{P_{\theta}(y)} \right] \end{aligned}$$

-ve entropy
(doesn't depend on θ)

So, $\boxed{\arg \max_{\theta} \frac{1}{N} LL(\theta) = \arg \min_{\theta} KL(P^* || P_{\theta})}$

So when we do MLE, we are trying to get closer to the reality, if we have infinite data.

With ∞ data,
 $\lim_{N \rightarrow \infty} \frac{\#(Y=1)}{N} = P(Y=1|\theta^*)$
 $\uparrow$
 $P^*(y)$ under the "true" model
 $\text{Let } P(Y=y|\theta) \Rightarrow P_{\theta}(y)$

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MLE is a frequentist ~~est~~ estimation.

Our answer was 3/5 as there were 3 wins & 2 losses. What if # win = 30, # loss = 20? → We'll still have the same answer. MLE estimator can't convey/confident on how much data you have seen.

MAP

We can put a belief on anything (eg. I know from my heart that India will win). That's Bayesian. In frequentist, θ is constant, it's not random. In Bayesian,

$$P(\theta|D) = \frac{P(D|\theta) P(\theta)}{P(D)}$$

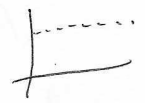
$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(\theta|D)$$

(after you observe some data, what do you believe about this)

If $P(\theta)$ is constant, then MAP reduces to

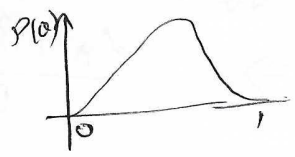
MLE:

$$\hat{\theta}_{MAP} = \hat{\theta}_{MLE}$$



Prior: for $P(\theta)$, we need some probability density function between 0 & 1.

distribution or some plot

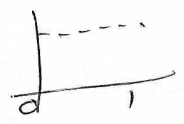


A useful prior:

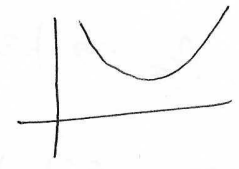
$$P(\theta) = \frac{\theta^{\beta_H - 1} (1-\theta)^{\beta_T - 1}}{B(\beta_H, \beta_T)} \sim \text{Beta}(\beta_H, \beta_T)$$

β_H, β_T hyperparameters.

- $\beta_H = \beta_T = 1$



$\beta_H = 0.6$
 $\beta_T = 0.6$



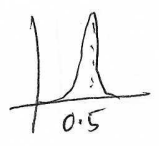
- $\beta_H = 2.5$
 $\beta_T = 2.5$



$\beta_H = 0.1$
 $\beta_T = 4.0$



- $\beta_H = 5.0$
 $\beta_T = 5.0$



$\beta_H = 4.0$
 $\beta_T = 0.6$



$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} p(\theta | D) \propto p(D | \theta) p(\theta) \\ &\propto \theta^{\alpha_H} (1-\theta)^{\alpha_T} \theta^{\beta_H-1} (1-\theta)^{\beta_T-1} \\ &\propto \frac{\theta^{\alpha_H + \beta_H - 1} (1-\theta)^{\alpha_T + \beta_T - 1}}{\text{constant}} \\ &\propto \text{Beta}(\alpha_H + \beta_H, \alpha_T + \beta_T)\end{aligned}$$

Conjugate prior

Now let's derive the MAP:

$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} p(\theta | D) = \theta^{\alpha_H} (1-\theta)^{\alpha_T} \frac{\theta^{\beta_H-1} (1-\theta)^{\beta_T-1}}{\text{constant}} \\ &= \arg \max_{\theta} \log \left[\theta^{\alpha_H} (1-\theta)^{\alpha_T} \frac{\theta^{\beta_H-1} (1-\theta)^{\beta_T-1}}{\text{constant}} \right]\end{aligned}$$

$$\frac{d}{d\theta} \log p(\theta | D) = \frac{1}{\theta} [\alpha_H \log \theta + \alpha_T \log (1-\theta) + (\beta_H-1) \log \theta + (\beta_T-1) \log (1-\theta)] = 0$$

$$\Rightarrow \frac{d}{d\theta} [(\alpha_H + \beta_H - 1) \log \theta + (\alpha_T + \beta_T - 1) \log (1-\theta)] = 0$$

$$\Rightarrow \frac{\alpha_H + \beta_H - 1}{\theta} - \frac{\alpha_T + \beta_T - 1}{1-\theta} = 0$$

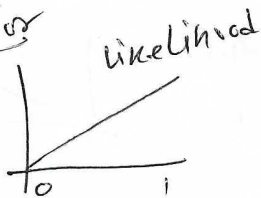
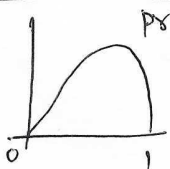
$$\Rightarrow \theta (\alpha_T + \beta_T + \alpha_H + \beta_H - 2) = \alpha_H + \beta_H - 1$$

$$\Rightarrow \hat{\theta}_{MAP} = \frac{\alpha_H + \beta_H - 1}{\alpha_T + \beta_T + \alpha_H + \beta_H - 2}$$

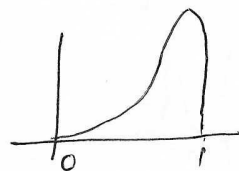
($\beta_H, \beta_T \rightarrow$ pseudo counts — reason ??)

As you see infinite data ($n \rightarrow \infty$), your prior is forgotten!
So MAP converges to MLE with ∞ data. But for small sample size, prior is important.

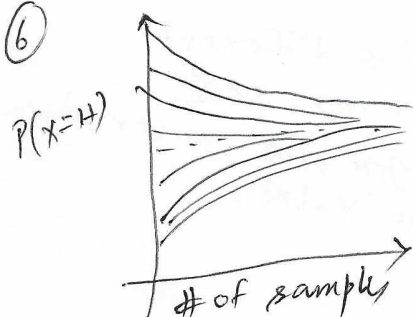
Effect of prior



posterior



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You start with different priors, but as you see more data, you are converging to the same value.

Frequentist tool $\Rightarrow$ MLE
Bayesian tool $\Rightarrow$ MAP

MAP vs. Full Bayesian.

We are interested in $P(\theta|D)$

By Bayes' rule: $P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$

Now $P(D) = \int P(D|\theta)P(\theta) d\theta$ $\leftarrow$ marginal likelihood or evidence.
 $\leftarrow$ This integral is the root of difficulty.

Consider every possible parameter setting θ , see how well it explains the dataset D , weight it by how plausible θ was a-priori, and add everything up.

So Bayes must account for all possible models, not just the best one.

Why MAP avoids the hard part?

$$\theta_{MAP} = \arg \max_{\theta} P(D|\theta)P(\theta)$$

No integral, no normalization, no averaging.
MAP doesn't care about $P(D)$ at all!!

In coin toss, θ is a number, $P(D|\theta)$ is simple, $P(\theta)$ is simple, & the integral has a closed form.

Now imagine complex ML model (like Neural Networks). So the integral becomes:

$$\int_{\mathbb{R}^{10^6}} P(D|\theta)P(\theta) d\theta$$

Integration over million dimensional space $\rightarrow$ no closed form soln.

MAP $\rightarrow$ point estimate

Full Bayesian $\rightarrow$ full posterior distribution.

MAP asks $\rightarrow$ What single parameter value is most plausible after seeing the data?

Full Bayesian inference $\rightarrow$ What is the entire distribution of plausible parameters?

Bayes does not ask for best θ ; Bayes asks for the contribution of all θ . MAP ignores them.

Often Bayesian involves approximating the integral.